<u>Confocal Imaging of Entire Whole Mount Adipose Tissues on Leica TCS SP8 DLS</u>

Jake W. Willows, Magdalena Blaszkiewicz, Kristy L. Townsend
ktownsendlab.com

Oct. 24th, 2020

Setup

1. Start-up the confocal microscope and Leica (LASX) software.
2. Make sure slide has no finger prints or debris on coverslip.
    a. Use a kimwipe and 70% EtOH to clean.
3. Select appropriate stage insert for your slide.
4. Place slide coverslip down in slide holder on the stage.
5. Use eyepiece to get tissue into focus at 10X objective magnification.
6. On LASX turn on white light laser, or appropriate laser line depending on your fluorophores.
7. Adjust excitation wavelengths to appropriately match the tissue fluorophores.
8. Turn on filters for each desired fluorescence channel. HyD filters are the most sensitive and typically yield better results for dim fluorescence signal with the highest contrast. The other available filter type is PMT. <u>NOTE:</u> HyDs can shut down if sensors become over saturated. Occurs often if there are areas in the tissue that are much more intensely stained than others. Make sure to scan brightest areas at finalized setting to be sure that HyD filters do not shut off.
    a. Alternatively, use z-compensation to adjust laser intensity depending on each z-plane. We do not typically do this.
9. After turning on desired filters turn on BrightR for the HyD filters by selecting the pulldown tab with 'standard' on the filter selected. Select 'BrightR'.
10. Select box for gating (leave default settings) to eliminate chances of reflected light interference.
11. Narrow the emission filter to the emission spectra peak of tissue fluorophore, and approximately 10nm away from the excitation peak wavelength.
    a. ex) if imaging 488nm fluorophore, set start of emission filter to 498nm at minimum.
12. Repeat for desired channels.
13. Adjust laser intensity and gain per channel for optimal imaging
14. Fluorophores can be excited and imaged all at once or sequentially. Sequentially eliminates channel bleed through but greatly increases time to image. Not all channel combinations will have bleed through, but many do. Sequential imaging is highly recommended.
    a. The best way to determine if sequential scanning is necessary is by scanning simultaneously with all desired channels at once (each set for optimal laser

intensity, gain, etc.) and then while live scanning the tissue turn off only one of the lasers. If signal decreases in any channel besides the one that was turned off then that channel was bleeding through and your imaging should be done sequentially in regards to the channels it bleeds into.

    i. UV light bleeds through practically all channels and needs to be imaged separately.
    ii. 488nm and 594nm are typically fine to be scanned simultaneously. 488nm will bleed into 594 though if using a very high laser intensity.
    iii. 488nm and 555nm require sequential scanning always.

  b. Imaging sequentially between frames by selecting the box for 'between frames'

15. Select desired resolution in the left-hand corner.
    a. 512x512 looks fine if only looking at the entire max projection. 1024x1024 allows for resolution of small filaments though. 720x720 is a good middle ground if imaging a whole depot and quantifying.
    b. Greater resolution causes increased imaging times and file sizes.
16. Select desired scan speed, typically 400-600Hz.
17. Select number of line averaging desired. Increased line averaging reduces image gain to an extent. For low gain setting 3-6 lines averaged is more than sufficient.
18. Select 'bidirectional acquisition' to speed up total acquisition time.
19. 10% overlap between tiles.

Set Tiling Area

20. Once all laser settings have been optimized click on the "stage overview" icon in the top left to start to map out tissue borders.
21. Find the edge of the tissue and scan the image repeatedly following the edge all the way around the entire tissue to determine where all borders are.
22. Next select the "polygon" icon at the bottom center of the screen. Trace the outline of the entire tissue border as close to the tissue edges as possible.

Set Z-Steps

23. Go to middle of tissue and set focal point at about center mass.
24. Use z-step control to find the 'highest point' or most positively in the z-direction' of the tissue (z-step start point.) Scan various areas of the tissue specifically the edges and very center of the tissue adjusting max height as you go to make sure no part of the tissue exceeds the top z-depth boundary.
25. Repeat for the 'lowest point' or most negative point in z-direction (z-step end).
    a. It is important to note that the stage can only accommodate a z-stack of -250um to +250um with highest signal being at the focal point (0um.)

b. This means that tissues exceeding 500um thick cannot be imaged without excluding data.

26. Select z-step size. Typically, 10-16um steps yield great results for max projecting data. If wanting to make a 3D image with clear z-resolution a 3um z-step is preferred. Smaller z-steps exclude less data with the caveat of greatly increasing image acquisition time.
    a. Larger z-step sizes can cause concentric rings in the final maximum projection image. This effect is compounded by increased tissue thickness.
27. Start imaging
    a. After the first tile is imaged the software will calculate total time required to image the tissue. Is typically anywhere from 13hr for a healthy tissue to >35hrs for obese tissue.
    b. To reduce acquisition time try:
        i. Increase z-step size
        ii. Increase scan speed
        iii. Decrease resolution
        iv. Decrease line averaging
        v. Decrease optical zoom

Processing Images

28. After entire tissue has been imaged find the merged tile file and open it in the processing tab.
29. Select projections and under the pulldown tab select maximum projection. Apply. It will take several minutes.
30. Open the tiled maximum projection image, adjust the LUT table if nescessary, and save the project.
31. Export the individual tile images as tiffs into a folder.
32. Export the merged images as tiffs into a separate folder.
33. Export the merged maximum projections into a third folder.
34. Proceed to quantification. See link https://github.com/ktownsendlab/blaszkiewicz_et_all-2019